Management Science

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To cite this article:
Si Li, Xintong Zhan (2019) Product Market Threats and Stock Crash Risk. Management Science 65(9):4011-4031.
https://doi.org/10.1287/mnsc.2017.3016

informs
http://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 65, No. 9, September 2019, pp. 4011–4031
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Product Market Threats and Stock Crash Risk

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Received: February 25, 2016
Revised: May 15, 2017; October 11, 2017
Accepted: November 18, 2017
Published Online in Articles in Advance: July 11, 2018

https://doi.org/10.1287/mnsc.2017.3016

Abstract. This paper examines the effect of product market threats on firms' stock crash risk. Competitive pressure from the product market aggravates managers' incentive to withhold negative information. When negative information is accumulated to a tipping point, the accumulated information comes out all at once and leads to an abrupt and large decline in stock price. Using a measure of product market threats, our regressions find that firms facing more threats are more prone to stock crashes. This result is confirmed by an instrumental variable analysis and a difference-in-difference analysis with an exogenous shock to market competition.

History: Accepted by Neng Wang, finance.

Funding: S. Li acknowledges financial support from the Ira Gluskin Fellowship in Finance and the Social Sciences and Humanities Research Council of Canada. X. Zhan acknowledges generous financial support of the Research Grant Council of the Hong Kong Special Administrative Region, China [Project CUHK 14501115].

Supplemental Material: The online appendix is available at https://doi.org/10.1287/mnsc.2017.3016.

Keywords: market competition • product market threats • stock crash risk

1. Introduction

Large fluctuations in stock prices, especially large sudden drops in asset prices, are one of the primary interests of investors and regulators. Recent literature emphasizes individual companies' downside tail risk or crash risk. For instance, Bali et al. (2014, p. 208) mention that "the tail risk of the individual stock will [also] matter for the tail risk of the underdiversified portfolio." Kelly and Jiang (2014) find that common variation in the tail risk of individual firms has strong predictive power for aggregate market returns. There is also a growing interest in how firm characteristics affect individual firms' crash risk, which is defined as large, negative, market-adjusted returns on individual stocks. This literature provides theoretical and empirical evidence that a firm experiences heightened crash risk when firm-specific bad news is concealed and accumulated to a threshold and then gets released all at once. In this paper, we examine an overlooked determinant of individual firms' crash risk: product market competitive threats.

How will product market competition affect firm-level stock crash risk? Market competition could increase a firm's crash risk because competitive threats aggravate managerial career concerns, which are an important consideration in financial reporting and bad news hoarding (Graham et al. 2005, Kothari et al. 2009). Career concerns cause managers to withhold bad news because bad news disclosure may lead to reduced executive compensation and/or job termination (Kim 1999, Kothari et al. 2009, Nagar 1999). Managerial career concerns are heightened when greater competition erodes firm profitability and increases CEO turnover rates (DeFond and Park 1999, Fee and Hadlock 2000). Consequently, in response to more intense competition, managers have stronger incentives to conceal bad news. In addition, competition increases uncertainty, which may cause managers to become overconfident about firm prospects. As a result, managers may have a greater tendency to disclose positive information rather than negative information.

When negative information is stockpiled, the share price of the firm will be artificially inflated. When the information accumulates to a point where the cost of hiding information becomes too high or where managers' behavioral bias is corrected, the accumulated bad news comes out all at once and leads to an abrupt and large decline in stock price (Jin and Myers 2006, Kothari et al. 2009). As a result, the "hidden negative information" hypothesis suggests a positive link between market competition and crash risk.

In contrast, competitive pressures from product market may decrease crash risk for several reasons. First, information revealed by competitors could convey information about the firm. Hence, it may be harder to hide negative information because there are additional sources that investors can draw on to infer information. Less information hoarding thus leads to less surprise for investors and thus lower crash risk. Second, competitive threats decrease agency costs by forcing managers to exert more efforts and to manage the firm more efficiently (Hart 1983, Schmidt 1997).

Lower agency costs thus reduce crash risk (Kim et al. 2011a, b). Third, a firm may act more conservatively on increased threats from rival firms (Hoberg et al. 2014, Dhaliwal et al. 2014). The financially conservative reaction can provide financial flexibility to firms, allowing them to better deal with unfavorable situations and react more aggressively to competitive threats when the threats materialize. Firms are thus less likely to experience large stock price declines. Taken together, the "competitors' information disclosure," "agency," and "financial conservatism" views suggest a negative relation between competition and stock crashes.

We empirically investigate the impact of product market competition on crash risk to shed light on the above competing views. We measure stock crash risk using two variables: the first variable, following Hutton et al. (2009) and Kim et al. (2011a, b), is a dummy variable equal to one for a firm-year if the firm has experienced any firm-specific weekly return more than three standard deviations below the mean return over the entire fiscal year. The second variable is the negative conditional return skewness, following Chen et al. (2001).

Our main competition measure is *Fluidity*, constructed by Hoberg et al. (2014). *Fluidity* measures similarity between a firm's products and the aggregate changes in the competitors' products. When *Fluidity* is greater, the firm's products are closer to its competitors' and thus the competitive threat is greater. We also use the competition measure constructed by Li et al. (2013), which is the number of occurrences of competition-related words per 1,000 total words in the firm's annual financial reports (10-K). Differing from the traditional industry-level competition measures such as the Herfindahl index and industry concentration ratios, the competition measures we use are firm specific and can better capture firm-level effects.

Our baseline probit and OLS regressions show a significant and positive relation between competitive threats and crash risk. We then use instrumental variable and natural experiment approaches to mitigate potential endogeneity of the competition variable. Specifically, we follow Xu (2012) and use import tariff and foreign exchange rate as instrumental variables for the competitive pressure from foreign rivals. We then use large reductions in U.S. import tariff rates as exogenous shocks to foreign competition. Import tariff forms an important entry barrier for foreign rivals and reduces the pressure from foreign competition (Fresard 2010, Valta 2012). A higher exchange rate (dollar amount of foreign currency in U.S. dollars) makes foreign goods cheaper in U.S. dollars and encourages imports. We obtain consistent results from the instrumental variable (IV) regression and the difference-in-difference analysis based on the exogenous shocks.

Our empirical results suggest that the relation between competition and crash risk is consistent with the information-withholding hypothesis. To investigate the information hiding channels and the triggers of stock crashes, we conduct an array of empirical tests. First, we study the readability of a firm's annual financial reports (10-Ks). Decreased readability in 10-Ks has been documented to be related to adverse news hiding (Li 2008, Rogers et al. 2014). Our empirical results show that a firm's 10-Ks become more difficult to read given stronger competitive pressure. We further find that on more severe threats from competitors, management earnings forecasts become more positive and less negative compared to analyst consensus. This suggests that managers may hide bad news and inflate investor expectations by issuing more positive earnings forecasts.

The artificially inflated expectations, however, cannot be sustained because adverse information will accumulate to a tipping point (at which the cost of hiding information becomes prohibitively high). Then the hidden adverse information will get released and the stock crashes. To study what triggers crashes, we determine all of the S&P 500 firms that experienced crashes in 2009. We then manually search the news to identify the reasons for crashes. In about half of the crashes for which we can identify the reason, the trigger was related to inflated profits, which lead to stock crashes when realized profits are found to be disappointing (often through earnings restatements or conference calls).[1] These case studies confirm our argument that stock will crash when adverse information, accumulated to a tipping point, is disclosed all at once.

To provide more systematic evidence, we conduct two additional empirical tests on how the triggers of stock crashes are linked to market competition. First, we interact the competition measure with past and contemporaneous firm profitability, return on assets (Roa). We find that higher lagged Roa and lower concurrent Roa are associated with a greater impact of competition on crash. The result suggests that lagged Roa is probably inflated (a bubble is built up) and later leads to crash (the bubble bursts). Second, we examine the stocks that crash during earnings conference call weeks, conditional on different levels of competitive threats. We find that firms facing more competition are significantly more prone to crashes than firms that face less competition during the conference call weeks. In the high-competition group, 7% of the stocks experience crashes during the conference call weeks, while the percentage is only 3% in the low-competition group. Furthermore, stock crashes mostly occur during the conference call weeks, with much fewer crashes occurring outside of the weeks of conference calls. This further verifies that crashes are often triggered when managers' earnings forecasts are below investors' expectations.

Collectively, our results suggest that managers are more likely to hide bad news when facing more threats from their rivals and the sudden release of adverse information leads to the stock crash. The results also suggest that the unexpected downward earnings adjustment is the main driving force underlying stock crashes. To better understand managers' incentives, we conduct various subsample analyses. The subsample results confirm that the positive link between competition and crash is stronger when the costs for managers to release bad news are higher (i.e., in firms with weaker market positions and tighter financial constraints), and when the costs of hiding bad news are lower (i.e., in firms operating in a more opaque information environment).

Our contributions to the literature are threefold. First, this paper adds to the stock crash literature by examining the role of competitive threats from a firm's product market. The existing literature on stock crash is mostly focused on the effects of stock market characteristics on crashes (Chen et al. 2001, Hong and Stein 2003, and Huang and Wang 2009, among others). There is a growing literature on how firm characteristics affect the crash risk of individual firms. For example, Hutton et al. (2009) and Jin and Myers (2006) examine the effect of information on crash risk and find that information transparency is related to less crash risk at the individual stock level. Kim et al. (2011a) find that tax avoidance facilitates managerial rent extraction and bad news hoarding activities, which lead to stock price crashes when the accumulated hidden bad news crosses a tipping point and comes out all at once. Kim et al. (2011b) argue that managerial option incentives induce managerial opportunism such as hiding bad news, which is related to higher stock crash risk. We add to this literature by showing that product market competition could affect firm-specific crash risk.

Second, we contribute to the literature on product market competition by showing that an unintended consequence of competition could be large stock price declines. The literature on product market competition consists of two views. The first view holds that competition is good: reducing agency costs (Hart 1983), increasing firm productivity (Syverson 2011), and encouraging innovation to some extent (Aghion et al. 2005). The opposite view is that competition has adverse effects: facilitating imitation, discouraging innovation when competition is high (Aghion et al. 2005), and causing unethical behavior (Shleifer 2004). Our paper adds to this literature by showing that another unfavorable effect of market competition could be large share price declines.

Third, the literature has examined the mean and the variance effects of market competition on stock returns, and our paper complements this literature by identifying a significant third moment (skewness) effect. Regarding the mean effect, extant studies find that competitive pressures enhance firm productivity (Galdon-Sanchez and Schmitz 2002, Syverson 2011). Hou and Robinson (2006) find that firms in more competitive industries earn higher returns, and they interpret the finding as these firms being riskier and thus investors commanding higher expected returns. On the variance effect, Irvine and Pontiff (2009) find that the increase in idiosyncratic volatility over the past 40 years is attributable to the more intense competition over the years. Valta (2012) finds that competition increases cash flow risk and thus default risk, which is reflected in the increased cost of bank debt. Ang et al. (2006) quantify a significant risk premium for investors bearing downside risk, and Chang et al. (2013) find a significant market skewness risk premium. Our results of competition being related to higher skewness risk suggest that the higher returns in firms in more competitive industries could be partially attributed to skewness risk premium.

2. How Does Competition Affect Crash Risk?

How does product market competition affect firm-level stock crash risk? Competitive pressures from product market may increase crash risk. Large share price declines or crashes occur because of the sudden release of unexpected bad news (Dye 2001, Healy and Palepu 2001, Verrecchia 2001). According to the theoretical model in Jin and Myers (2006), an average investor, possessing only public information, cannot anticipate the adverse private information. Stockpiled bad news, when suddenly released to investors, will lead to a large surprise and a stock crash.[2] This is confirmed by the empirical work on stock crash risk (e.g., Hutton et al. 2009; Kim et al. 2011a, b). One of the major factors causing bad news hiding arises from managerial career concerns because bad news disclosure may lead to reduced executive compensation and/or job termination (Kim 1999, Kothari et al. 2009, Nagar 1999). Product market competition aggravates such career concerns because it erodes firm profitability and increases CEO turnover rates (DeFond and Park 1999, Fee and Hadlock 2000). Consequently, in response to fiercer competition, managers have stronger incentives to conceal bad news, and the stock is more likely to experience crashes.

The key result of the corporate disclosure theory (Dye 1985, 2001; Verrecchia 1983, 2001) states that the equilibrium is a threshold equilibrium. That is, firms disclose only the information that is more favorable than the threshold and withhold the information that is less favorable than the threshold. The primary factors impeding information disclosure involve managers trading off the benefits with the costs of withholding information. The important consideration is the benefits and costs arising from managerial career concerns (Kim 1999, Nagar 1999).[3] Kothari et al. (2009) present the evidence that firm managers, because of career concerns, withhold bad news. Similarly, Ali et al. (2015) find that managers withhold unfavorable news out of career concerns. Managers' career and reputation concerns become more severe when competition weakens a firm's market share and decreases its ability to earn profits. Diminishing firm prospects increase the probability that the manager's compensation package shrinks and/or his job is terminated. Existing empirical evidence shows that more-competitive product markets display greater management turnover rates (DeFond and Park 1999, Fee and Hadlock 2000). As a result, competition increases the cost of disclosing bad news (e.g., reduced compensation packages and increased likelihood of job termination). The cost of hiding bad news, such as litigation risk (Skinner 1994), however, does not seem to change substantially when competition is intensified. As a result, managers facing more competition have stronger incentives to hide adverse information.

When negative information is stockpiled, share price will possibly be artificially inflated. Bad news, nevertheless, may finally become public in the long run. Based on the model in Jin and Myers (2006), managers have an abandonment option of releasing accumulated bad news. The option is exercised if the manager is forced to absorb a sufficiently long period of firm-specific bad news and cannot absorb any additional news. According to Kothari et al. (2009), managers stop accumulating the adverse information when it becomes too costly or difficult to do so. Therefore, when negative firm-specific information is accumulated to that threshold, the accumulated bad news comes out all at once and leads to an abrupt and large decline in stock price. In all, the "hidden negative information" hypothesis suggests a positive link between market competition and crash risk.

In contrast, competitive pressures from product market may decrease crash risk. There could be a few reasons for this. First, competition could also lead to less hidden (negative) information because a signal about one firm may reveal information about its competitors. If there is any common component among competitors, then information revealed about one firm would be informative about other firms. Hence, competitors' disclosures can be a source from which investors can infer negative information about their firm, making it harder for managers to hide bad news. Competition, therefore, may lead to decreased crash risk.

Second, competitive threats decrease agency costs, and lower agency costs reduce crash risk (Kim et al. 2011a, b). Threats from competitors force managers to exert more efforts and to manage the firm more efficiently (Hart 1983, Schmidt 1997). This results in decreased agency costs, improved firm productivity, and thus a lower possibility that stock will drop significantly in a sudden manner.

Third, a firm may act more conservatively on increased product market threats from rival firms. For example, Hoberg et al. (2014) show that firms facing competitive threats adopt more conservative financial policies, with reduced payouts and higher cash balances. Dhaliwal et al. (2014) show that firms under competitive pressures are more conservative in financial reporting. The financially conservative reaction to competitive threats can provide financial flexibility to firms, allowing them to better deal with unfavorable situations and react more aggressively to competitive threats when the threats materialize. Firms are thus less likely to experience large stock price declines. Consistent with this idea, Kim and Zhang (2016) show that accounting conservatism is associated with lower stock crash risk.

In sum, the "hidden negative information" hypothesis suggests a positive link between market competition and crash risk, while the "competitors' information disclosure," "agency," and "financial conservatism" views suggest a negative relation between competition and stock crashes. These competing hypotheses leave the relation between competition and firm-specific crash risk an unanswered empirical question, which we will investigate below.

3. Variable Measurements

3.1. Measuring Firm-Specific Crash Risk

We use weekly CRSP (Center for Research in Security Prices) stock returns to construct annual firm-specific crash risk. Closely following Kim et al. (2011a, b), for each firm year, we use weekly returns during the 12-month period ending three months after a firm's fiscal year end to construct the crash risk variables. The three-month lag ensures that the financial data are available to investors and reflected in stock prices when measuring crash risk (Kim et al. 2011a).[4] For each firm year, we require at least 26 weekly returns. We exclude observations with nonpositive book equity, with nonpositive total assets, or with fiscal year-end stock prices less than \$1. We also remove utility ($4000 \leq SIC \leq 4999$) and financial ($6000 \leq SIC \leq 6999$) firms because these industries are regulated and have different competition landscapes than other industries.

We then follow two steps to compute firm-specific crash risk. First, for each stock, we run the following regression to remove the impact of market returns and obtain firm-specific weekly returns:[5]

$$r_{i,t} = \alpha_i + \beta_{1i} r_{m,t-2} + \beta_{2i} r_{m,t-1} + \beta_{3i} r_{m,t} + \beta_{4i} r_{m,t+1} + \beta_{5i} r_{m,t+2} + \varepsilon_{it}, \quad (1)$$

where $r_{i,t}$ is the return on stock i in week t, and $r_{m,t}$ is the return on the CRSP value-weighted market index in week t. The lead and lag terms for market returns are included to remove the impact of nonsynchronous trading (Dimson 1979). Specifically, for infrequently traded stocks, the prices recorded at the end of a time period represent the outcome of a transaction that occurred earlier in or prior to the period. Dimson (1979) thus includes preceding, synchronous, and subsequent market returns to deal with the nonsynchronous trading problem. We calculate firm-specific weekly returns as the log residual return: $W_{i,t} = \ln(1 + \hat{\varepsilon}_{i,t})$.

The second step uses $W_{i,t}$ to construct two crash variables. The first variable, *Crash*, is equal to one if a firm experiences one or more crash weeks in a fiscal year, and zero otherwise. We define the crash week as the week in which the firm's weekly return $W_{i,t}$ is 3.2 standard deviations below the mean firm-specific weekly returns over the entire fiscal year for this firm. Following Kim et al. (2011a, b), 3.2 is chosen so that the crash events account for 0.07% of frequency in the normal distribution. That is, if firm-specific returns $W_{i,t}$ are normally distributed, one would expect to observe 0.07% of the sample observations crashing in any week.[6]

Our second crash variable is the negative conditional return skewness, *Ncskew*. Following Chen et al. (2001), we calculate *Ncskew* for a given firm year by taking the negative of the third moment of firm-specific weekly returns $W_{i,t}$ for each sample year and then dividing it by the standard deviation of $W_{i,t}$ raised to the third power:

$$Ncskew_{is} = -\frac{n(n-1)^{3/2} \sum W_{it}^3}{(n-1)(n-2)(\sum W_{it}^2)^{3/2}}, \qquad (2)$$

where n is the number of observations on weekly returns during year s. Intuitively, *Ncskew* measures left-tail thickness, scaled by the standard deviation of the returns. The scaling allows for comparing stocks with different volatilities. The minus sign in front of the equation allows us to interpret the variable in a natural way that an increase in *Ncskew* corresponds to a stock having a more left-skewed distribution and thus being more prone to crash.

Using weekly returns to construct the crash variables is a trade-off. As noted in Jin and Myers (2006), using short-horizon returns is an advantage in estimating the moments such as skewness. However, the use of high-frequency returns could introduce noise or oddly shaped residual returns. For example, a large, negative, firm-specific return in a particular week might reverse in the next week and not really be the crash predicted by firm fundamentals. We therefore check our results using daily and monthly returns and find robust results.

3.2. Measuring Competition Pressure

Our main measure of firm-specific competitive pressure is the *Fluidity* variable developed by Hoberg et al. (2014) and available from the Hoberg–Phillips Data Library. *Fluidity* captures the similarity between a firm's products and the changes of the products made by competitors in the firm's product market. If a firm's products overlap more with the dynamic changes of the rivals' products (i.e., higher *Fluidity*), then the firm faces more competitive threat. We also construct the ranking of product market fluidity, *r_Fluidity.* In each fiscal year, we obtain the decile rank of the sample firms based on their *Fluidity* levels and scale the ranks to be $0.1, 0.2, \ldots, 1$. Then we merge the fluidity data with our crash risk data. By requiring nonmissing fluidity and control variables, we are left with a sample of 27,955 unique firm-years, covering 4,759 publicly traded U.S. firms over the period from 1998 to 2009.[7]

In addition to *Fluidity* and its transformation *r_Fluidity*, we use *Pctcomp* and *r_comp* (developed by Li et al. 2013) as alternative measures of firm-specific competitive pressure. Li et al. (2013) count the number of references to competition in the firm's 10-K filings and develop two measures to capture the notion that more intense behavior from new and existing rivals diminishes a firm's ability to earn profits. *Pctcomp* is the number of occurrences of competition-related words per 1,000 total words in the 10-K, and reflects management's perceptions of the intensity of the competition they face. According to Li et al. (2013), their competition measures are correlated with existing industry-level measures of competition (such as the Herfindahl index) but capture something distinctly new. The *Pctcomp* measure has both across-industry and within-industry variations and is related to the firm's future rates of diminishing marginal returns. We merge the *Pctcomp* data (retrieved from Feng Li's website) with our sample and obtain 17,285 unique firm-year observations with nonmissing *Pctcomp*. We then calculate *r_comp*, the decile-ranked value of *Pctcomp*, based on our sample. *r_comp* is computed each year and scaled to be $0.1, 0.2, \ldots, 1$. The original sample with available *Pctcomp* ranges from 1995 to 2009. To be consistent with the fluidity sample, we constrain the final sample to 1998–2009.

In this paper, we focus on the above-mentioned competition variables rather than the traditional ones such as the Herfindahl index (HHI) and industry concentration ratios (CR). First, *Fluidity* and *Pctcomp* are firm-level variables and capture firm-specific information that is unavailable in the industry-level measures such as HHI and CR. Second, *Fluidity* and *Pctcomp* are forward-looking and incorporate the dynamic actions of a firm's rivals, while HHI and CR are static and based on historical information on firm market shares (Hoberg et al. 2014, Li et al. 2013). Third, *Fluidity* and *Pctcomp* capture competition not only from public firms, but also potentially from private firms. Ali et al. (2009) find that failing to consider private firms when

Table 1. Summary Statistics

Variables	No. of obs.	Mean	SD	$p25$	Median	$p75$
Dependent variable						
$Crash_t$	27,995	0.184	0.388	0	0	0
$Ncskew_t$	27,995	0.018	0.813	−0.450	−0.017	0.425
Independent variable: Main						
$Fluidity_{t-1}$	27,995	0.406	0.2	0.255	0.373	0.523
Independent variable: Alternative						
$r\_Fluidity_{t-1}$	27,995	0.545	0.287	0.3	0.5	0.8
$Pctcomp_{t-1}$	17,285	0.546	0.448	0.225	0.403	0.733
r_comp_{t-1}	17,285	0.549	0.287	0.3	0.5	0.8
Control variables						
$Dturn_{t-1}$	27,995	0.008	0.106	−0.025	0.002	0.036
$Sigma_{t-1}$	27,995	0.067	0.034	0.042	0.06	0.085
Ret_{t-1}	27,995	−0.281	0.308	−0.356	−0.175	−0.085
$Size_{t-1}$	27,995	5.703	1.886	4.329	5.599	6.965
Mb_{t-1}	27,995	2.099	1.702	1.121	1.538	2.371
Lev_{t-1}	27,995	0.139	0.158	0.003	0.085	0.222
Roa_{t-1}	27,995	0.079	0.179	0.047	0.114	0.17

Notes. The table reports the descriptive statistics for crash risk, product market threats, and control variables. The measures of crash risk are *Crash* and *Ncskew.* The main measure for product market threats is *Fluidity* developed by Hoberg et al. (2014). The alternative competition measures include *r_Fluidity, pctcomp,* and *r_comp.* The sample contains 27,995 unique firm-years for 4,759 publicly traded U.S. firms over the period from 1998 to 2009. Variable definitions are in Online Appendix Table A1. All variables are winsorized at 1% and 99%.

calculating market shares will result in poor proxies for the actual industry concentration. Hoberg et al. (2014) show that *Fluidity*, though measuring threats from primarily public firms through 10-K, is significantly correlated with the competitive threats from private entrepreneurial firms. *Pctcomp* reflects management's perceptions of the intensity of the competition they face and thus incorporates competition from many sources such as public and private firms and potential new entrants (Li et al. 2013).

3.3. Summary Statistics

Table 1 presents the summary statistics of the variables used in the analysis. After deleting the observations with missing control variables, the sample contains 27,995 unique firm-years from 4,759 publicly traded U.S. firms from 1998 to 2009. All of the variables are within reasonable ranges and in line with the statistics reported in the literature (Chen et al. 2001; Kim et al. 2011a, b; Hoberg et al. 2014). On average, 18.4% of the sample firm-year observations experience one or more crash weeks each year. The average *Ncskew* is positive (0.018), indicating that the sample firms' returns are negatively skewed on average. The median *Ncskew* is negative at −0.017 and lower than the mean value, suggesting that some sample firms experience extremely low returns.

We also compute the correlations between the variables and report the results in Online Appendix Table A2. The two crash risk measures are highly correlated, with a correlation of 0.64. The main competition measure, *Fluidity*, is positively related to both measures of crash risk. The correlation is 0.03 for the *Crash* indicator and 0.05 for *Ncskew*. The other three competition measures are also positively correlated with the crash variables. These correlations suggest that firms facing more competitive threats are also more prone to stock crash. Further, the correlations between the control variables show that these variables do not present any collinearity problem.

4. Empirical Results

4.1. Effect of Competitive Threats on Crash Risk—Baseline Results

In this section, we empirically test the relation between competitive threats and firm-specific crash risk. We run probit (with the dependent variable being the *Crash* indicator) and OLS (with the dependent variable being *Ncskew*) regressions, linking the four competition measures (*Fluidity*, *r_Fluidity*, *Pctcomp*, and *r_comp*) in year $t-1$ to firms' crash risk in year t. Following the literature on the determinants of crash risk (Chen et al. 2001; Hutton et al. 2009; Kim et al. 2011a, b), we control for a variety of variables (which are explained in detail later in this section and also defined in Online Appendix Table A1). We also control for industry fixed effects (FE) and year fixed effects.[8] We adjust standard errors for heteroskedasticity and cluster the standard errors at the firm level.

In panel A of Table 2, we report the marginal effects of competitive threats (evaluated at mean) on the *Crash* indicator. Consistent with the positive correlations

Table 2. Probit and OLS Regressions

Variables	(1)	(2)	(3)	(4)
Panel A: Dependent variable = $Crash_t$				
Independent variable: Main				
$Fluidity_{t-1}$	0.066***			
	[0.015]			
Independent variables: Alternative				
$r\_Fluidity_{t-1}$		0.043***		
		[0.010]		
$Pctcomp_{t-1}$			0.022***	
			[0.008]	
r_comp_{t-1}				0.030***
				[0.011]
Control variables				
$Dturn_{t-1}$	0.093***	0.093***	0.086***	0.086***
	[0.023]	[0.023]	[0.032]	[0.032]
$Ncskew_{t-1}$	0.003	0.003	−0.003	−0.003
	[0.003]	[0.003]	[0.004]	[0.004]
$Sigma_{t-1}$	0.947***	0.951***	2.079***	2.056***
	[0.305]	[0.305]	[0.424]	[0.423]
Ret_{t-1}	0.131***	0.132***	0.258***	0.255***
	[0.032]	[0.032]	[0.048]	[0.048]
$Size_{t-1}$	0.008***	0.008***	0.012***	0.012***
	[0.002]	[0.002]	[0.002]	[0.002]
Mb_{t-1}	0.007***	0.007***	0.008***	0.008***
	[0.002]	[0.002]	[0.002]	[0.002]
Lev_{t-1}	−0.047**	−0.047**	−0.037	−0.037
	[0.019]	[0.019]	[0.023]	[0.024]
Roa_{t-1}	0.146***	0.140***	0.229***	0.230***
	[0.018]	[0.017]	[0.033]	[0.033]
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
No. of obs.	27,995	27,995	17,284	17,284
Pseudo R^2	0.0273	0.0272	0.0317	0.0317

observed in Table 1, all of the four competition measures have statistically significant and positive coefficients. We also evaluate the economic significance of the marginal effects. The coefficient on *Fluidity* is 0.066. Given a one-standard-deviation change (0.2) in *Fluidity*, the probability of *Crash* changes by $0.066 \times 0.2 = 0.0132$ or 1.32%. This is compared to the average *Crash* frequency of 18.4% (out of all sample observations) and the standard deviation of *Crash* of 38.8%. Later we show, in column (3) of Table 3, that the instrumental variable regression gives a much larger marginal effect estimate of 0.344, which generates greater economic magnitude of 6.88% ($=0.344 \times 0.2$). The coefficients on other competition variables have similar economic magnitude. Overall, the results suggest that competitive threats faced by a firm could predict the probability that the firm will experience large stock price declines in the following year.

In panel B of Table 2, we report the OLS regression results with *Ncskew* as the dependent variable. The results again give positive and significant coefficients on all of the four competition measures. The coefficient on *Fluidity* is 0.166 and translates into a 0.033 (0.166×0.2) change in the *Ncskew* when *Fluidity* changes by one standard deviation of 0.2. The magnitude is large compared to the mean *Ncskew* of 0.018. Similarly, the economic magnitude is 0.03 (0.104×0.287) for *r_Fluidity*. In addition, a one-standard-deviation (0.448) increase in *Pctcomp* is associated with an increase of 0.016 (0.036×0.448) in *Ncskew*, and the economic magnitude on *r_comp* is the same at 0.016 (0.055×0.287).

Following prior literature (Chen et al. 2001, Hutton et al. 2009), we control for the variables that may affect stock crash risk. We first control for stock characteristics. We include *Dturn*, the year-to-year change in average monthly share turnover, where monthly share turnover is the monthly trading volume divided by the total number of shares outstanding during the month. Trading volume proxies for the differences in opinions, which may cause stock crashes according to Chen et al. (2001). In addition, as return skewness may be serially correlated, we include lagged skewness in all of the regressions. We also include *Sigma*, standard deviation of firm-specific weekly returns during the fiscal year,

Table 2. (Continued)

Variables	(1)	(2)	(3)	(4)
Panel B: Dependent variable = $Ncskew_t$				
Independent variable: Main				
$Fluidity_{t-1}$	0.166***			
	[0.033]			
Independent variables: Alternative				
$r\_Fluidity_{t-1}$		0.104***		
		[0.022]		
$Pctcomp_{t-1}$			0.036**	
			[0.016]	
r_comp_{t-1}				0.055**
				[0.022]
Control variables				
$Dturn_{t-1}$	0.298***	0.298***	0.251***	0.251***
	[0.048]	[0.048]	[0.063]	[0.063]
$Ncskew_{t-1}$	0.008	0.008	0.004	0.004
	[0.007]	[0.007]	[0.008]	[0.008]
$Sigma_{t-1}$	4.633***	4.665***	5.720***	5.679***
	[0.600]	[0.600]	[0.776]	[0.776]
Ret_{t-1}	0.502***	0.505***	0.620***	0.615***
	[0.059]	[0.059]	[0.080]	[0.080]
$Size_{t-1}$	0.055***	0.055***	0.057***	0.057***
	[0.004]	[0.004]	[0.004]	[0.004]
Mb_{t-1}	0.037***	0.037***	0.041***	0.040***
	[0.003]	[0.003]	[0.005]	[0.005]
Lev_{t-1}	−0.286***	−0.287***	−0.292***	−0.293***
	[0.038]	[0.038]	[0.047]	[0.047]
Roa_{t-1}	0.325***	0.312***	0.437***	0.438***
	[0.035]	[0.035]	[0.063]	[0.063]
Constant	−0.747***	−0.742***	−0.774***	−0.795***
	[0.044]	[0.044]	[0.057]	[0.058]
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
No. of obs.	27,995	27,995	17,285	17,285
Adjusted R^2	0.057	0.057	0.056	0.056

Notes. This table reports the regression results on the effect of competition pressure on crash risk. Panel A shows the marginal effects of competitive threats on the *Crash* indicator from the probit regressions. Panel B shows the OLS regression results on the effect of competitive threats on *Ncskew*. Column (1) presents the result using *Fluidity* as the main independent variable. Columns (2)–(4) employ alternative measures of competitive pressure. Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets.
*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

to account for skewness's being correlated with volatility. *Ret*, average firm-specific weekly returns (percentage) over the fiscal year period, is included because returns predict skewness. We then control for firm characteristics, including firm *Size* (log of total assets), *Mb* (market-to-book ratio, defined as market value of equity divided by book value of equity), *Lev* (leverage, equal to total debt divided by market value of assets), and *Roa* (profitability, measured as income before extraordinary items divided by total assets). These variables are associated with firm performance and thus firm-specific crash risk.

The coefficients on the control variables are consistent with previous studies. We find that share turnover and volatility are positively related to crash risk, suggesting that firms that have experienced a surge in turnover or volatility are more crash-prone. Firm size is positively related to crash risk, consistent with the finding in the literature that crash is more likely to occur in large-cap firms (Chen et al. 2001, Hutton et al. 2009). Growth stocks are prone to crash possibly because the prices of growth stocks are more likely to be inflated. Leverage is associated with lower crash risk, which may reflect the fact that more stable, less crash-prone firms are more willing or able to establish higher levels of indebtedness (Hutton et al. 2009). We include additional control variables, liquidity and return kurtosis, in robustness analysis. The unreported results show that both liquidity and kurtosis are positively related to crash risk and the sign and magnitude of the coefficient on the competition variable remain similar.

We also find that crash risk is positively related to past stock returns and lagged firm profitability *Roa*. When we further explore our information hypothesis in Section 5, we find that crash risk is negatively related to contemporaneous profitability (Table 6, panel A). The relations could be explained by the stochastic bubble model in Chen et al. (2001); that is, high profitability and high stock returns indicate a "bubble" built up, so that there is a larger price drop when the bubble pops. Such a bubble could be due to the fact that firms hide bad information, resulting in inflated profits and share price. The inflated share price then leads to stock crash when bad information is released and profits are corrected downward. In unreported analysis, we attempt different measures of profitability (including return on equity, net income divided by sales, and alternative measures of *Roa*), and find similar results.

Overall, our regression results reveal a positive relation between competitive pressure and firm crash risk. The evidence shows that the "information concealing" hypothesis dominates the competing hypotheses. Endogeneity issues, however, may exist. For example, unobserved shocks (such as macroeconomic shocks that we cannot fully control for) could affect both product market competition and stock market crash risk. Reverse causality may also be an issue. For instance, firms with better stock performance (and thus lower crash risk) may gain competitive advantages in product markets by obtaining financing at a lower cost. The next sections attempt to address the endogeneity issue, using an instrumental variable (IV) approach and a natural experiment.

4.2. Robustness

4.2.1. Evidence from the Instrumental Variable Regressions. To alleviate potential endogeneity from omitted variables or reverse causality, we first use the IV method. Specifically, we follow Xu (2012) and use import tariff (*Tariff*) and foreign exchange rate (*Exrt*) (both at industry level) as IVs for the competition variable *Fluidity*. These two IVs satisfy the relevance condition because they are correlated with the endogenous variable: market competition. According to Bernard et al. (2007) and Tybout (2003), import tariff forms an important entry barrier for foreign rivals and reduces the pressure from import competition. Exchange rate (dollar amount of foreign currency in U.S. dollars) is positively correlated with foreign competition, as a higher exchange rate makes foreign goods cheaper in U.S. dollars and encourages import (Xu 2012). In addition, both IVs satisfy the exclusion condition because they are arguably not related to firm-level crash risk through any channels other than the competition channel.

To obtain the tariff data, for each three-digit SIC industry year, we calculate the ad valorem tariff rate as the duties collected by the U.S. customs divided by the free-on-board value of imports. The information on the duties and the value of imports is collected from Feenstra (1996), Feenstra et al. (2002), and Schott (2010), and covers the U.S. manufacturing firms for the years until 2005. We obtain the raw exchange rate data from the International Financial Statistics of the International Monetary Fund (IMF), and convert the raw rate into the real rate using the exchanging countries' consumer price indices from the IMF. Following Xu (2012), we construct the industry-level (three-digit SIC) foreign exchange rate as the source-weighted average of exchange rates across all exporting countries. The weights are the share of each exporting country in the three-digit SIC industry in 1997. We choose 1997 as the base year as our sample begins in 1998. The weights are fixed over time because according to Xu (2012), most industries have stable import shares by country. Our IV regressions are based on the manufacturing firms ($2000 \le \text{SIC} \le 3999$) over the period of 1998–2005.

We conduct two tests to show that these manufacturing firms do not differ significantly from our main full sample. First, the summary statistics of these manufacturing firms (reported in Online Appendix Table A3) are similar to those of the main sample. Second, the empirical results tabulated in Table 3 show that the probit and OLS regression results using these manufacturing firms (columns (1) and (2)) are similar to those using the full sample.

Our main IV results are reported in Table 3. In the first stage of the IV regression, we regress *Fluidity* on tariffs, exchange rates, firm controls, and fixed effects in column (5), and find that both tariff and exchange rates are significant in predicting *Fluidity*. The negative sign on tariff indicates that higher tariff rates reduce competition from foreign rivals, and the positive sign on the exchange rate indicates that higher exchange rates make foreign goods cheaper and thus facilitate the entry of foreign rivals. The F-statistic of 25.77 rejects the null that the instruments are jointly zero, suggesting that the IVs are significantly related to market competition and the relevance condition for IVs is satisfied.

We then replace *Fluidity* with the predicted value of *Fluidity* from the first-stage regression, and generate the IV estimates in columns (3) and (4) of Table 3. The Hansen J-statistics, which test the overidentifying restrictions and zero correlation between the instruments and the error term, are insignificant with p-values of 0.81 and 0.24. This indicates that the instruments are appropriately uncorrelated with the disturbance process of the model and thus satisfy the exclusion condition. We also conduct the Anderson–Rubin (AR) test. A significant AR chi-square statistic indicates

Table 3. IV Regressions

	Probit	OLS	IV second-stage estimation		IV first-stage estimation
Variables	(1) Dep. var. = $Crash_t$	(2) Dep. var. = $Ncskew_t$	(3) Dep. var. = $Crash_t$	(4) Dep. var. = $Ncskew_t$	(5) Dep. var. = $Fluidity_{t-1}$
$Fluidity_{t-1}$	0.041**	0.109**	0.344**	0.642**	
	[0.020]	[0.044]	[0.148]	[0.310]	
$Tariff_{t-1}$					−2.731***
					[0.381]
$Exrt_{t-1}$					0.093**
					[0.044]
$Dturn_{t-1}$	0.079**	0.258***	0.108***	0.291***	−0.105***
	[0.031]	[0.065]	[0.039]	[0.082]	[0.014]
$Ncskew_{t-1}$	0.001	0.002	0.001	0.004	−0.003
	[0.004]	[0.009]	[0.005]	[0.011]	[0.002]
$Sigma_{t-1}$	1.010**	4.989***	−0.328	2.987	4.859***
	[0.401]	[0.794]	[0.907]	[1.853]	[0.252]
Ret_{t-1}	0.116***	0.491***	0.027	0.349**	0.356***
	[0.042]	[0.079]	[0.075]	[0.147]	[0.024]
$Size_{t-1}$	0.009***	0.055***	0.003	0.053***	0.025***
	[0.002]	[0.005]	[0.005]	[0.009]	[0.002]
Mb_{t-1}	0.006***	0.031***	0.000	0.023***	0.016***
	[0.002]	[0.004]	[0.004]	[0.007]	[0.002]
Lev_{t-1}	−0.088***	−0.338***	−0.026	−0.246***	−0.182***
	[0.026]	[0.055]	[0.042]	[0.088]	[0.019]
Roa_{t-1}	0.110***	0.258***	0.211***	0.435***	−0.288***
	[0.021]	[0.045]	[0.049]	[0.103]	[0.016]
Constant		−0.708***		−0.732***	0.122***
		[0.057]		[0.091]	[0.030]
Year FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
No. of obs.	15,846	15,846	11,955	11,955	11,955
Pseudo/adj. R^2	0.02	0.05		0.045	0.509
P-value of Hansen's J-statistic			0.81	0.24	
Anderson–Rubin (A-R) test (robust to weak instruments)		χ^2 statistic (p-value)	8.53 (0.01)	5.96 (0.05)	
Test: IVs jointly equal to zero			F-statistic (p-value)		25.77 (0.00)

Notes. This table presents the two-stage least squares (2SLS) regression results on the effect of competition on crash risk, using the instrumental variable approach. The instrumental variables for competition include the import tariff rate and the real effective exchange rate at the three-digit SIC level. Because these IVs are available for manufacturing firms (SIC 2000-3999) only, the regressions in this table are conducted on manufacturing firms (the summary statistics of this sample are provided in Online Appendix Table A3). The table presents the regression results. Columns (1) and (2) are the probit and OLS regressions based on the manufacturing firms, and these results are presented to compare with the 2SLS results in columns (3) and (4). Columns (3) and (4) present the second-stage regression results, and column (5) reports the first-stage estimation. The estimates in columns (1) and (3) are marginal effects. Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets (clustering standard errors at the industry level gives similar results).

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

that the effect of the endogenous variable *Fluidity* in the model is indeed significantly different from zero, and that our IV estimates are robust to weak instruments.[9]

In all, our probit, OLS, and IV regressions show that competitive pressure leads to higher crash risk. The results are consistent with the negative information withholding hypothesis that greater competitive threats are associated with more withholding of unfavorable information, which in turn leads to stock crashes.

4.2.2. Evidence from a Natural Experiment. To further confirm the causal relation between competition and crash risk, we utilize the exogenous reduction in tariff rates as a natural experiment.[10] The substantial reductions in import tariffs initiated by U.S. authorities over the past 30 years represent exogenous shocks that change the competitive landscape of the related sectors (Fresard 2010, Valta 2012). Similar to the requirement of the instrumental variables, an ideal natural experiment should satisfy both relevance and exclusion conditions. Regarding the relevance condition, the literature shows that import tariffs are an important fraction of trade costs (Anderson and van Wincoop 2004) and the reduction of import tariff lowers trade barriers and intensifies foreign competition (Tybout 2003). Most of the recent tariff changes occurred under the auspices of international institutions such as the General Agreement on Tariff and Trade (GATT) and more recently the World Trade Organization (WTO). We can reasonably argue that the rules of these international institutions are uncorrelated with domestic firms' stock crash risk and thus justify the exclusion condition.

We employ annual import tariff data for U.S. manufacturing industries from Feenstra (1996), Feenstra et al. (2002), and Schott (2010). Because a new classification scheme is used to record U.S. industry-level imports and exports starting in 1989, we restrict the sample to be after 1989. For each industry (three-digit SIC) year, we compute the ad valorem tariff rate as the duties collected by the U.S. customs divided by the free-on-board value of imports. To capture meaningful shocks on product market competition, we focus on large tariff reductions. Following Fresard (2010) and Valta (2012), we define a negative shock to import tariffs in year t to be one if the tariff reduction is more than three times of the median reduction in the same industry over the entire sample period. Furthermore, to ensure that the tariff reduction events are nontransitory, we exclude the events followed by large tariff increases within the next two years (Fresard 2010). We also exclude the events with the ex ante tariff rates smaller than 1% because import restrictions with such low tariff rates are likely minimal.[11]

Based on the above criteria, 57 tariff reductions are classified as exogenous shocks during 1990–2005. These reductions give 695 unique treatment firms and 1,456 firm-year observations. We match these observations with untreated firms by year, firm size, Tobin's Q, market leverage, and Roa. The matched sample contains 620 unique firms and 1,456 firm-year observations. We conduct a balance test to verify that treated and matched firms do not significantly differ from each other in any of our matching dimensions (reported in Online Appendix Table A4).

After obtaining a valid matching sample, we compute the treatment effect using the difference-in-differences (DID) approach. According to Roberts and Whited (2012), the key assumption for consistency of the DID estimator is the parallel trend assumption. Specifically, this assumption requires any trends in outcomes (i.e., crash variables) for the treatment and control groups prior to treatment (i.e., tariff reduction) to be the same. To check that our data satisfy the assumption, we perform statistical tests of the mean differences in the changes of the outcome variables between the treated group and the matched group. The tests in panel A of Table 4 find that the pre-event growth of the

Table 4. Evidence from Natural Experiments

Panel A: Pre-event trends in firm crash risk for treated and matched firms

		Average change from $t-3$ to $t-2$	Diff. between treated and matched (t-stat)	Average change from $t-2$ to $t-1$	Diff. between treated and matched (t-stat)
Crash indicator	Treated	0.010	0.013 (0.97)	0.009	0.004 (0.38)
	Matched	−0.003		0.005	
Ncskew	Treated	−0.038	0.004 (0.66)	0.001	0.0015 (0.07)
	Matched	−0.042		−0.0005	

Panel B: Nearest-neighbor matching results

	Treated firms		Nearest-neighbor matched firms	Difference in difference	
	No. of obs.	Mean difference (Post − Pre)	Mean difference (Post − Pre)	Average treatment effect (ATT)	T-stat
Crash indicator	1,456	0.020	−0.003	0.023**	1.96
Ncskew	1,456	0.033	−0.088	0.121***	4.41

Notes. This table reports the impact of reduced tariff rates on stock crash risk using the difference-in-differences approach. The sample contains 1,456 unique firm-year observations that experience exogenous reduction in industry tariff rate. The firms are matched with control firms by year, size, market-to-book, leverage, and Roa, based on the nearest-neighbor matching method. After matching, the sample contains 695 unique treatment firms and 620 unique control firms from 1990 to 2005. The balance test results that show that the matched firms do not differ significantly from the treatment firms are contained in Online Appendix Table A4. Panel A of this table checks the pre-event parallel trend assumption. The average treatment effects (ATT) based on nearest-neighbor matching are tabulated in panel B. Variable definitions are in Online Appendix Table A1. All variables are winsorized at 1% and 99%.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

outcome variables does not differ significantly across the two groups.

After verifying the matching sample validity and the parallel trend assumption, in panel B of Table 4, we conduct the DID analysis using the treated sample and the nearest-neighbor matched counterfactuals. The increase in *Crash* ratio in treated firms is 0.023 higher than that in control firms, and the difference is significant at the 5% level. The number, 2.3%, can be compared with the mean *Crash* of 18.3% before the tariff cut (see statistics on manufacturing firms in Online Appendix Table A3). The average treatment effect (ATT) of tariff cut on *Ncskew* is 0.121, and its t-statistic is 4.41. The effect is economically significant, considering that the mean *Ncskew* for the treated firms before the shock is −0.016 on average (Online Appendix Table A3). Overall, by exploiting the exogenous shock in import tariff rates, we establish causality from competitive threats to firm crash risk.

5. Exploring the Information Channel

5.1. Bad News Withholding—Channels and Triggers

The results thus far establish a robust and positive causality from competition to stock crashes. We argue that the positive relation is due to the fact that product market threats aggravate bad news hiding and the stock crashes when bad news is released all at once. To further support our argument, in this section, we conduct a series of empirical tests to investigate the information hiding channels and the triggers of stock crashes.

5.1.1. Financial Report Readability. First, financial reports are one of the major means through which managers communicate information to investors. Managers can "bury" adverse information by obscuring financial reports. Based on Li (2008), managers strategically use less readable and longer annual reports to make information less transparent and to hide adverse information from investors (pp. 225, 245). Previous studies also use financial report readability measures to proxy for managers' adverse news hoarding behavior (e.g., Rogers et al. 2014). Or, financial reports may simply become more obscure because competition increases the uncertainty and makes the industry and/or firm more complicated to analyze. To investigate whether competitive threats lead to less readable financial reports, we examine the relation between competition and financial report readability. Li (2008) provides *Fog* (defined as $0.4 \times$ words per sentence $+ 0.4 \times$ percent of complex words used in the annual report, where complex words are words with three syllables or more), *NegFlesch* ($1.015 \times$ words per sentence $+ 84.6 \times$ syllables per word $- 206.835$), and *Kincaid* ($0.39 \times$ words per sentence $+ 11.8 \times$ syllables per word $- 15.59$) as measures of financial report readability, with higher values indicating less readable financial report. Li (2008) also provides *Length*, the logarithm of the number of words in the annual report. A longer report is more deterring and more difficult to read. Our results find that more competitive threats is associated with reduced readability of financial reports (greater *Fog*, *NegFlesch*, *Kincaid*, and *Length*) in both the concurrent year and the following year (Table 5, panel A). This suggests that when firms face more threats from competitors, managers are more likely to withhold adverse information through obscure financial reports.

5.1.2. Management Earnings Forecasts. The analysis on annual reports provides suggestive evidence on information hiding on more intense competition. To supply more direct evidence, we examine another important means by which managers communicate information to investors: management earnings forecasts. In particular, we obtain managers' forecasts of quarterly earnings per share (EPS) from First Call, and compare these forecasts with analyst consensus for the same period. We categorize managers' earnings forecasts as positive (negative) if managers' forecasts are higher (lower) than analyst consensus. If managers provide more positive and less negative earnings forecasts relative to analysts' forecasts, these managers are possibly hiding unfavorable firm-specific information or fail in disclosing unfavorable information perhaps because they are overconfident about the firm. We construct three annual variables: the number of positive forecasts for the year (*Pos_num*), the number of negative forecasts (*Neg_num*), and the percentage of negative forecasts out of the total number of management earnings forecasts (*Neg_pct*). We then regress these variables on product market threats. The results show that when the product market becomes more competitive, managers provide more positive and less negative forecasts in both the same year and the following year (Table 5, panel B). This is consistent with managers' increased tendency to withhold bad news and inflate investors' expectations when facing more threats from rivals.

5.1.3. How Crashes Are Triggered. The artificially inflated expectations, however, cannot be sustained because adverse information will accumulate to a tipping point (i.e., the bubble will burst ultimately). The hidden information then comes out all at once and leads to crashes. To study what triggers crashes, we extensively search the news to identify the possible reasons for stock crashes. Specifically, we first determine all of the S&P 500 firms that experienced crashes in 2009.[12] We then carefully read the relevant news and are able to identify the triggering reasons for 19 crashes.[13] Out of these 19 stock crashes, three are triggered by accounting restatements or fraud detections, in which managers inflate earnings. Six crashes

Table 5. Managers' Bad News Withholding

Panel A: Effect of competitive threats on financial report readability

	Dep. var. at time $t-1$ and controls at time $t-2$				Dep. var. at time t and controls at time $t-1$			
	(1) Fog_{t-1}	(2) $NegFlesch_{t-1}$	(3) $Kincaid_{t-1}$	(4) $Length_{t-1}$	(5) Fog_t	(6) $NegFlesch_t$	(7) $Kincaid_t$	(8) $Length_t$
$Fluidity_{t-1}$	0.948***	3.403***	1.128***	0.802***	0.986***	3.264***	1.163***	0.781***
	[0.123]	[0.336]	[0.118]	[0.041]	[0.110]	[0.303]	[0.106]	[0.037]
Size	0.045***	0.238***	0.077***	0.125***	0.050***	0.215***	0.084***	0.127***
	[0.016]	[0.040]	[0.015]	[0.005]	[0.015]	[0.036]	[0.014]	[0.004]
Mb	−0.014	−0.072**	−0.005	0.008**	−0.004	−0.052*	0.004	0.009**
	[0.011]	[0.030]	[0.010]	[0.004]	[0.010]	[0.027]	[0.010]	[0.003]
Lev	−0.250	−2.263***	−0.454***	0.322***	−0.087	−1.910***	−0.333**	0.318***
	[0.162]	[0.418]	[0.154]	[0.053]	[0.145]	[0.354]	[0.136]	[0.044]
Roa	−0.380***	−0.533*	−0.483***	−0.310***	−0.475***	−0.732**	−0.566***	−0.277***
	[0.118]	[0.323]	[0.114]	[0.041]	[0.122]	[0.315]	[0.116]	[0.036]
Constant	19.080***	−22.896***	14.909***	9.212***	19.022***	−22.235***	14.896***	8.975***
	[0.147]	[0.313]	[0.133]	[0.037]	[0.147]	[0.291]	[0.130]	[0.036]
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	18,060	18,060	18,060	17,979	24,189	24,189	24,189	24,063
Adjusted R^2	0.041	0.148	0.062	0.168	0.038	0.164	0.059	0.155

Panel B: Effect of competitive threats on management earnings forecasts

	Dep. var. at time $t-1$ and controls at time $t-2$			Dep. var. at time t and controls at time $t-1$		
	(1) Poisson Pos_num_{t-1}	(2) Poisson Neg_num_{t-1}	(3) Tobit Neg_pct_{t-1}	(4) Poisson Pos_num_t	(5) Poisson Neg_num_t	(6) Tobit Neg_pct_t
$Fluidity_{t-1}$	0.324**	−0.09	−11.31*	0.222	−0.097*	−10.97*
	[0.149]	[0.060]	[6.752]	[0.148]	[0.056]	[6.109]
Size	0.030*	0.093***	0.555	0.004	0.086***	1.540**
	[0.016]	[0.007]	[0.771]	[0.016]	[0.006]	[0.735]
Mb	0.038	−0.033***	−1.616	−0.023	−0.001	1.794*
	[0.027]	[0.011]	[1.296]	[0.025]	[0.007]	[0.968]
Lev	−0.005	−0.438***	−7.072	0.309	−0.453***	−21.79**
	[0.236]	[0.089]	[10.29]	[0.213]	[0.079]	[9.214]
Roa	2.678***	0.210*	−75.91***	0.419*	0.559***	5.427
	[0.326]	[0.118]	[13.83]	[0.244]	[0.087]	[9.951]
Constant	−0.549	−1.113***		−0.307	−0.202	
	[0.482]	[0.425]		[0.902]	[0.621]	
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	5,693	5,693	5,693	6,405	6,405	6,405
Pseudo R^2	0.0627	0.0628	0.00974	0.0458	0.0624	0.00819

Notes. This table presents the impact of competition on managers' bad news withholding behavior. In panel A, *Fog*, *NegFlesch*, and *Kincaid* are measures of financial report readability, with higher values indicating less-readable financial reports. *Length* is the logarithm of the number of words in the annual report. A longer report is more deterring and more difficult to read. In panel B, we obtain managers' forecasts of quarterly earnings per share (EPS) from First Call, and compare these forecasts with analyst consensus for the same period. We categorize managers' earnings forecasts as positive (negative) if managers' forecasts are higher (lower) than analyst consensus. *Pos_num* (*Neg_num*) is the number of managers' positive (negative) forecasts for the year. *Neg_pct* is the percentage of managers' negative forecasts out of the total number of management earnings forecasts for the year. In panel B, the Poisson model is used in columns (1), (2), (4), and (5), and the Tobit model is used in columns (3) and (6). Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

are triggered by disappointing firm performance. Such disappointing performance, unexpected by investors, is disclosed by managers during earnings conference calls, or caused by earnings announcements that miss the preset targets. The remaining triggers for stock crashes include random shocks unexpected by the market.[14] Overall, half of the crashes in S&P 500 firms in 2009 are related to inflated profits, which later lead to stock crashes when realized profits are found to be disappointing.

To provide more systematic evidence, we conduct two empirical tests on how the triggers of crashes are linked to product market competition. First, the above case studies suggest that crashes could be triggered when bad news is disclosed as unexpected declining profits. To investigate this, we interact the fluidity measure with past and contemporaneous firm profitability variables. As we hypothesize in the paper, stock will crash when profits are first inflated (i.e., a bubble is formed) and then found to be disappointing (i.e., the bubble bursts). We therefore expect that the interaction term between competition and past firm profits has a positive effect on crash risk; that is, greater past profits indicate more information withheld or a greater likelihood of a bubble formed, which will be more likely to lead to crashes later. We also expect that the interaction term between competition and current profits will negatively impact crash risk; that is, lower current profits indicate a burst bubble and a resulting crash.

The results in panel A of Table 6 are consistent with the above conjectures. The first two columns find that the impact of competition on crash risk is absent unless lagged profits are high, consistent with crashes being related to inflated earnings. The evidence also echoes the positive coefficients on the control variables, lagged stock returns and lagged profitability, in Table 2. Columns (3) and (4) in panel A of Table 6 show that the effect of competition on crash risk is stronger if contemporaneous profits are lower. Taken together, previously inflated profits trigger stock crashes when the profits are revealed to be worse than expected.

Second, our news search results find that crashes could be triggered when managers' earnings forecasts are lower than investors' expectations during earnings conference calls. Panel B of Table 5 presents the evidence that managers facing more competitive threats are more likely to provide opportunistically or optimistically positive earnings forecasts. Such earnings guidance inflates investors' expectations. As the major way for managers to communicate with investors, earnings conference calls provide an important channel through which bad news is released. As a result, we expect that when bad news is accumulated to the tipping point, the earnings guidance provided by managers in their conference calls is more likely to contain a downward correction of investors' beliefs. Such corrections, if substantial, may lead to stock crashes. To investigate this conjecture, we examine the stocks that crash during the conference call weeks, conditional on different levels of competitive threats these firms face.

In panel B of Table 6, we find that consistent with our conjecture, the proportion of the firms that experience stock crashes during the conference call weeks increases monotonically in the intensity of market competition. In the high-competition group, 7% of the stocks experience crashes during the conference call weeks, while the percentage is only 3% in the low-competition group. We then examine how the intensity of stock crashes changes over the cycle of earnings conference calls. Figure 1 shows that firms facing more competitive threats (top quintile) are significantly more prone to stock crashes than firms that face little competition (bottom quintile) during the conference call weeks. This echoes the evidence in panel B of Table 5 that managers of the firms with greater competitive threats tend to give more-positive earnings forecasts, which are related to a higher likelihood of stock crashes. Firms with fewer competitive threats (all else equal), in contrast, are less likely to have inflated

Table 6. Release of Adverse Information

Panel A: Effect of competitive threats on crash risk, conditional on past and contemporaneous Roa

Variables	(1) $Crash_t$	(2) $Ncskew_t$	(3) $Crash_t$	(4) $Ncskew_t$
$Fluidity_{t-1}$	−0.007	−0.014	0.099***	0.236***
	[0.022]	[0.036]	[0.020]	[0.045]
$High\ Roa_{t-1}$		0.002	0.009	
	[0.012]	[0.019]		
$Fluidity_{t-1} \times High\ Roa_{t-1}$	0.061**	0.101**		
	[0.027]	[0.044]		
$High\ Roa_t$			−0.022*	−0.034
			[0.024]	[0.053]
$Fluidity_{t-1} \times High\ Roa_t$			−0.020*	−0.054**
			[0.011]	[0.023]
$Controls_{t-1}$	Yes	Yes	Yes	Yes
Constant		−0.674***		−0.652***
		[0.045]		[0.048]
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
No. of obs.	27,995	27,995	27,955	27,955
Pseudo/adj. R^2	0.0268	0.062	0.0287	0.058

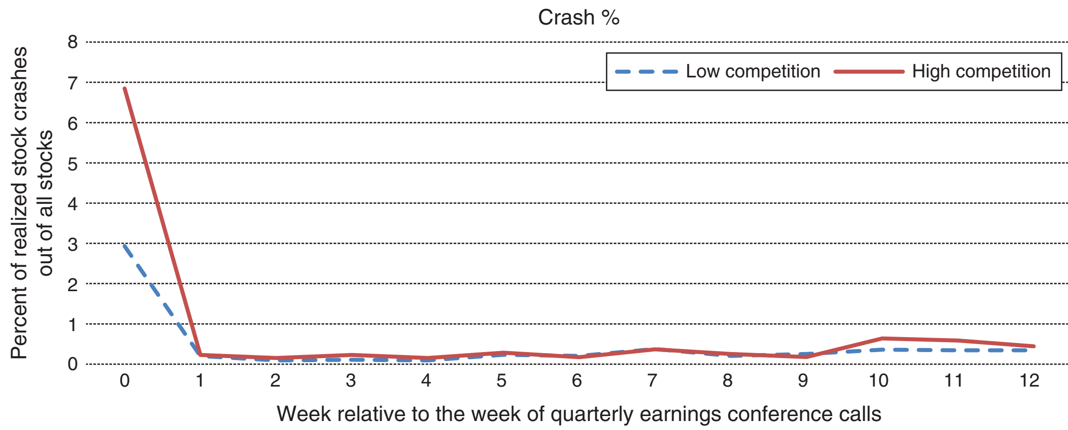
Panel B: Percentage of firms that experience crashes during earnings conference call weeks, by competition quintiles

	Competition quintile				
	Low	2	3	4	High
Crash (%)	2.93	4.25	5.06	5.52	6.85

Notes. This table investigates what triggers stock crashes and how the triggering is related to competition. Panel A reports the impact of competition on crash risk conditional on lagged and contemporaneous profitability (Roa). *High Roa* is an indicator variable equal to one if the firm's Roa is greater than the sample median, and zero otherwise. Panel B reports the percentage of the firms that experience crashes in the week of the quarterly conference calls, conditional on the competition level. Sample firms are classified into five groups each year, based on the competitive threats they face. Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure 1. (Color online) Frequency of Crashes By Week



Notes. This figure presents the percentage of firms that experience crashes by week. Week 0 is the week of quarterly earnings conference call. The solid (dashed) line is the firms facing higher (lower) competitive threats. The high-competition (low-competition) group consists of firms that face competitive threats in the top (bottom) quintile.

earnings, and thus are less prone to stock crashes. Furthermore, Figure 1 shows that stock crashes experienced by both groups are largely skewed during the conference call weeks, with fewer crashes occurring outside of the weeks of the conference calls. This verifies that earnings conference calls provide an important channel through which bad news is released.

Overall, in this section, we provide both anecdotal and systematic evidence on the channels and triggers of stock crashes. Our empirical evidence, combined with the theoretical literature on firm-specific crash risk (Jin and Myers 2006), highlights that information hoarding is the main mechanism through which competition leads to higher stock crash risk. The evidence also shows that the nature of the information being withheld is most likely earnings news. We, nevertheless, acknowledge that in addition to obscure financial reports and opportunistic earnings guidance, there may be other ways through which managers hide competition-induced diminishing prospects. In addition, the withheld adverse information could be disclosed through other channels, such as financial analysts acting as whistleblowers. Finally, managers may not be deliberately withholding bad information. Instead, they may withhold bad information unintentionally, possibly because they are overconfident about firm prospects.

5.1.4. A Further Investigation of the Profitability Channel. The previous sections show that the important channel through which competition affects crash risk is the news about firm profitability. In this section, we investigate the profitability channel in detail. First, we examine the difference in the timeliness of loss recognition between the low- and high-competition samples. We follow Basu (1997) and regress earnings on annual stock returns, negative return dummy, competition, and their interactions. We also control for industry and year fixed effects. The results in Table 7 show that the relation between earnings and stock returns is more pronounced when returns are negative, consistent with the literature (Basu 1997) that firms are timelier in recognizing losses than gains. More importantly, the coefficient on the interaction of competition, stock returns, and negative return dummy is equal to −0.090, negative and significant. This indicates that competition decreases the timeliness of loss recognition relative to profit recognition. That is, firms facing fiercer competition are more reluctant to recognize losses, providing suggestive evidence that these firms withhold negative news about earnings.

Second, we visually illustrate the earnings distribution changes before and after the crash, separately for the crash and noncrash samples. The density functions of the two samples in Figure 2 show that for the crash sample, earnings shift leftward after crashes. For the noncrash sample, the earnings distributions remain similar over time. This suggests that the earnings of crashed firms may be inflated before crashes, and the inflated earnings are corrected downward after crashes. The frequency distributions and cumulative distribution functions of the earnings of the crash versus noncrash samples (reported in Online Appendix Figure A1) confirm a correction of profitability after the crash.

5.2. Subsample Analysis

This section conducts various cross-sectional tests to better understand adverse information withholding. As mentioned in Section 2, managers' decision to withhold bad news and inflate investor expectations is a trade-off between the benefits and the costs of doing so. Therefore, the positive effect of competition on crash risk should be stronger when the benefits of bad

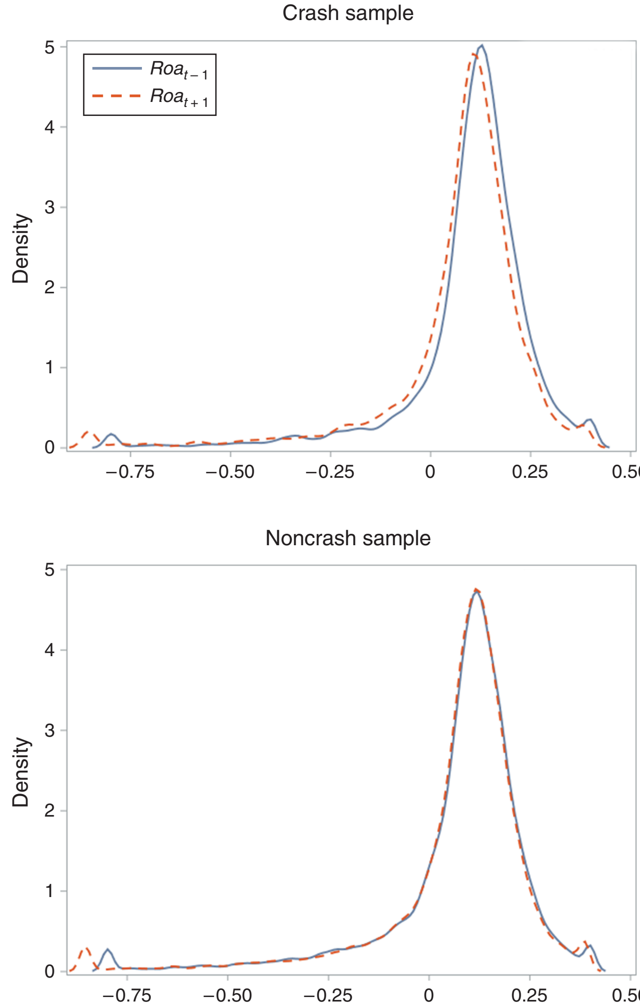
Table 7. Timely Loss Recognition, Conditional on Product Market Competition

	Dep. var. = EPS_t/P_{t-1}
$Fluidity_t \times NegD_t \times Aret_t$	−0.090**
	[0.039]
$Fluidity_t \times Aret_t$	−0.076***
	[0.014]
$Fluidity_t \times NegD_t$	−0.010
	[0.014]
$NegD_t \times Aret_t$	0.244***
	[0.020]
$Fluidity_t$	−0.117***
	[0.012]
$NegD_t$	0.000
	[0.006]
$Aret_t$	0.023***
	[0.007]
Industry FE	Yes
Year FE	Yes
No. of obs.	27,188
Adjusted R^2	0.146

Notes. This table investigates firms' timely loss recognition (Basu 1997), conditional on high and low product market competition. Following Basu (1997), the regression is contemporaneous and the dependent variable is measured as of the same fiscal year as the independent variables. The dependent variable is earnings per share (before extraordinary items) scaled by fiscal-year-beginning share price. *Aret*, annual stock returns, is measured as 12-month compound returns beginning nine months prior to fiscal year end. *NegD*, negative return indicator, is defined as one if annual stock returns are less than zero. The coefficient on *NegD* × *Aret* is the Basu (1997) coefficient, measuring the incremental timeliness of loss recognition in earnings relative to gains. Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure 2. (Color online) Earnings Distributions



Note. The figure plots the density functions of return on assets (*Roa*) distributions for one year before crash versus one year after crash, separately for crash and noncrash groups.

news hiding outweigh the costs to a larger extent, and weaker vice versa. Following this rationale, we examine three factors that could affect managers' incentives to withhold bad news: the firm's competitive position, financial condition, and information environment.

5.2.1. Competition Positions. Managers in firms with disadvantaged positions in competition may have a higher cost of releasing bad news when competition is intensified. Unlike the firms with favorable positions, weaker firms have little buffer against heightened competition. Releasing the same amount of adverse information could present a greater threat to weaker companies as well as their managers' careers. These managers may also be less competitive in the labor market. Consequently, managers in weaker firms face a higher cost of releasing bad news and are more likely to hide adverse news when there is more competition. We expect the effect of *Fluidity* on crash risk to be more pronounced in weaker firms.

To test the above argument, we define *Share* as one for firms with above-median market share (based on firm sales) in the three-digit SIC industry, and zero otherwise. We then include *Share* and its interaction with *Fluidity* in the regressions. We also examine the degree of concentration using the HHI by the text-based network industry classification (TNIC3HHI) (developed by Hoberg and Phillips 2010a, b and available from the Hoberg–Phillips Data Library). We define *Tnic* as an indicator equal to one if the firm has an above-median TNIC3HHI, and zero otherwise. We interact the indicator with *Fluidity*. Panel A of Table 8 finds negative coefficients on the interaction terms, indicating that the positive effect of competition on crash is stronger in firms with weaker competitive positions. In these firms, managers are more incentivized to hide adverse information because of more severe career concerns.

Table 8. Subsample Analysis

Panel A: Subsample analysis by competition positions

Variables	(1) $Crash_t$	(2) $Crash_t$	(3) $Ncskew_t$	(4) $Ncskew_t$
$Fluidity_{t-1}$	0.091***	0.099***	0.232***	0.230***
	[0.019]	[0.019]	[0.042]	[0.041]
$Share_{t-1}$	0.015		0.054*	
	[0.013]		[0.027]	
$Fluidity_{t-1} \times Share_{t-1}$	−0.068**		−0.168***	
	[0.028]		[0.059]	
$Tnic_{t-1}$		0.034***		0.063**
		[0.013]		[0.026]
$Fluidity_{t-1} \times Tnic_{t-1}$		−0.073**		−0.164***
		[0.028]		[0.059]
$Controls_{t-1}$	Yes	Yes	Yes	Yes
Constant			−0.780***	−0.781***
			[0.045]	[0.047]
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
No. of obs.	27,995	27,988	27,995	27,988
Pseudo/adj. R^2	0.0276	0.0276	0.057	0.057

Panel B: Subsample analysis by financial constraints

Variables	(1) $Crash_t$	(2) $Crash_t$	(3) $Crash_t$	(4) $Ncskew_t$	(5) $Ncskew_t$	(6) $Ncskew_t$
$Fluidity_{t-1}$	0.019	0.040**	0.071***	0.077*	0.097**	0.166***
	[0.020]	[0.020]	[0.017]	[0.042]	[0.042]	[0.038]
HP_{t-1}	−0.042***			−0.061**		
	[0.012]			[0.025]		
$Fluidity_{t-1} \times HP_{t-1}$	0.098***			0.172***		
	[0.025]			[0.054]		
WW_{t-1}		−0.026**			−0.054**	
		[0.013]			[0.027]	
$Fluidity_{t-1} \times WW_{t-1}$		0.050**			0.122**	
		[0.025]			[0.054]	
$Dividend_{t-1}$			0.005			−0.006
			[0.012]			[0.024]
$Fluidity_{t-1} \times Dividend_{t-1}$			−0.055*			−0.074
			[0.031]			[0.061]
$Controls_{t-1}$	Yes	Yes	Yes	Yes	Yes	Yes
Constant				−0.729***	−0.729***	−0.731***
				[0.047]	[0.050]	[0.045]
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	27,963	27,798	27,971	27,963	27,988	27,971
Pseudo/adj. R^2	0.0279	0.0276	0.0276	0.057	0.057	0.057

5.2.2. Financial Constraints. We then investigate the role of financial constraints. Financially constrained firms are vulnerable to aggressive pricing and production strategies adopted by competitors. Tighter financial constraints also imply difficulty in funding positive-NPV projects to gain advantages over rivals (Campello 2006). Constrained firms thus may hide negative information to obtain financing at a lower cost. Consequently, we expect that financially constrained firms have higher benefits of hiding bad information and are more affected by market competition than the unconstrained firms.

We rely on three measures of financial constraints: HP index (Hadlock and Pierce 2010), WW index (Whited and Wu 2006), and dividend paying indicator. For each fiscal year, a firm's HP Index is computed as $-0.737 \times Size - 0.043 \times Size^2 - 0.040 \times Age$, where *Size* is log(inflation-adjusted book assets), and *Age* is the current year minus the first year that the firm has a nonmissing stock price on Compustat. Higher HP index (that is, smaller and younger firms) indicates more financial constraints. The WW index is computed as $-0.091 \times CashFlow/AT - 0.062 \times Dividend + 0.021 \times Leverage - 0.044 \times Log(AT) + 0.102 \times IndustrySalesGrowth - 0.035 \times FirmSalesGrowth$, where

Table 8. (Continued)

Panel C: Subsample analysis by information asymmetry						
Variables	(1) $Crash_t$	(2) $Crash_t$	(3) $Crash_t$	(4) $Ncskew_t$	(5) $Ncskew_t$	(6) $Ncskew_t$
$Fluidity_{t-1}$	0.040**	0.107***	0.099***	0.127***	0.301***	0.192***
	[0.019]	[0.020]	[0.019]	[0.044]	[0.044]	[0.042]
$High\ Ret\_vol_{t-1}$	−0.018			0.008		
	[0.012]			[0.024]		
$Fluidity_{t-1} \times High\ Ret\_vol_{t-1}$	0.069***			0.143***		
	[0.025]			[0.055]		
$Large_{t-1}$		0.058***			0.211***	
		[0.012]			[0.024]	
$Fluidity_{t-1} \times Large_{t-1}$		−0.068***			−0.161***	
		[0.024]			[0.052]	
Old_{t-1}			0.021*			0.001
			[0.011]			[0.024]
$Fluidity_{t-1} \times Old_{t-1}$			−0.081***			−0.097*
			[0.025]			[0.054]
$Controls_{t-1}$	Yes	Yes	Yes	Yes	Yes	Yes
Constant				−0.520***	−0.472***	−0.736***
				[0.033]	[0.037]	[0.046]
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	27,995	27,995	27,995	27,995	27,995	27,995
Pseudo/adj. R^2	0.0273	0.0278	0.0279	0.056	0.054	0.057

Notes. This table presents the subsample analysis of the impact of competition on crash risk. Panel A presents the results partitioned by incumbent firms' competition positions: market share and product market Herfindahl index. *Share*/*Tnic* is an indicator variable that equals one if the firm has a market share/TNIC3HHI (Hoberg and Phillips 2010a, b) greater than the sample median, and zero otherwise. Panel B presents the results partitioned by firms' financial constraints: HP index (Hadlock and Pierce 2010), WW index (Whited and Wu 2006), and dividend paying indicator. *HP* (*WW*) is an indicator variable equal to one if the firm has an *HP* (*WW*) index higher than the sample median, and zero otherwise. *Dividend* is an indicator variable equal to one if the firm pays dividends, and zero otherwise. Panel C presents the results partitioned by firms' information asymmetry. *High Ret_vol* (*Large*, *Old*) is an indicator variable equal to one if the firm's stock return volatility (size, age) is higher than the sample median, and zero otherwise. Variable definitions are in Online Appendix Table A1. Standard errors adjusting for heteroskedasticity and within-firm clustering are in brackets. Control variables used in Table 2 are included in all of the estimations, but suppressed for expositional convenience. In panel C, the control variable sigma is excluded from columns (1) and (4) because *High Ret_vol* is in the models, and firm size is excluded from columns (2) and (5) because *Large* is in the models.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

CashFlow is operating cash flows, *Dividend* is an indicator that equals one if the firm pays cash dividends, *Leverage* is the ratio of long term debt to total assets, *AT* is total assets, and *IndustrySalesGrowth* is the average sales growth of all firms in the same three-digit SIC industry. Smaller, less profitable, and lower-growth firms have a higher WW index, which indicates more financial constraints. We categorize firms with the HP and WW indices above sample medians as more financially constrained. We also split the sample by whether a firm pays cash dividends because a dividend payer is less financially constrained (Denis and Sibilkov 2010). Panel B of Table 8 shows that the interaction terms between the two financial constraint indices and competition are positive and significant, and the interaction between the dividend payer indicator and competition is negative and significant. These results suggest that financial constraints aggravate managers' incentives to hide bad news and thus the effect of competitive pressure on crash risk becomes more pronounced.

5.2.3. Information Asymmetry. Firms operating in a more transparent information environment have a higher cost of hiding bad news (Kothari et al. 2009), suggesting that such firms are less likely to withhold bad news and thus less prone to crash risk. Furthermore, when the information environment is transparent, investors can better value the firm and anticipate firm-specific news; therefore, the share price is less likely to move out of surprise. Hutton et al. (2009) empirically document that opaque stocks are more likely to crash. We thus expect to see that the impact of product market threats on crash risk is more pronounced when the firm is less transparent. We split the sample into high and low information asymmetry groups, according to firm size, age, and volatility. *Large*, *Old*, and *High Ret_vol* are indicator variables equal to one if the firm's size, age, and stock return volatility are greater than the sample median, and zero otherwise. Panel C of Table 8 shows that the effect of competition on crash risk is stronger in smaller, younger, and more volatile firms. This result suggests that bad news hoarding caused by competitive threats is more severe in opaque firms, consistent with the cost of hiding information in these firms being lower.[15]

6. Discussion

We have thus far explored the mechanisms underlying the relation between product market threats and stock crashes. Our results provide the evidence that stock crashes are caused by initial suppression and subsequent release of bad news. Such information withholding could be a result of managers actively or deliberately withholding adverse information. We, nevertheless, acknowledge that other reasons could lead to hidden adverse information.

First, intense competition increases the riskiness of firms' business environment and the uncertainty about the future payoffs of potential projects (Fresard 2010, Valta 2012, Grenadier 2002, Raith 2003, Valta 2012). With increased uncertainty, managers and investors may become overly optimistic about firm prospects and incorrectly assess the value of their projects, focusing more on the upside (perhaps irrationally) than on the downside. That is, managers may be less likely to disclose bad information (unintentionally) because of overconfidence.

Second, it is possible that managers have no information advantage and discover the true underlying cash flow process in the same way as the investors. In this case, bad information may be hidden because the nature of the information generating process is such that it is less costly to gather positive information than to gather negative information.[16] Such an information gathering process may lead managers to withhold bad information unintentionally.

Third, competition may lead to less transparent information (for example, more obscure financial reports and less precise earnings forecasts) available to investors because competition increases the uncertainty and makes the industry and/or the firm more complicated to analyze. As a result, with an opaque information environment, it is more difficult for investors to value the firm and to anticipate firm-specific news, and therefore, the share price is more likely to move unexpectedly.

To summarize, the withholding of adverse information is not necessarily out of firm managers' opportunistic information manipulation but could be a result of unintentional behavior caused by behavioral bias or other factors. The results in this paper should be carefully interpreted with this point in mind.

7. Conclusions

This paper examines the effect of product market threats on firms' stock price crash risk, and finds that firms facing more threats are more prone to stock crashes. Market competition leads to bad news hoarding, which inflates investors' expectations and engenders stock price crashes when bad information is disclosed unexpectedly. Our results suggest that competition may adversely affect the financial market by increasing the likelihood of large stock price declines.

Acknowledgments

For providing the data used in the paper, the authors thank the Center for International Data at the University of California, Davis; the Hoberg–Phillips Data Library; and Feng Li. For helpful comments, the authors are grateful to department editor Neng Wang, one anonymous associate editor, two anonymous referees, Jie (Jay) Cao, Andrew Karolyi, Sophia Zhengzi Li, Chen Lin, Yaxuan Qi, Baolian Wang, and Lu Zhang. They also thank seminar participants at National Taiwan University, Peking University, the University of Regina, Zhejiang University, the 2014 Econometric Society China Meetings, the 22nd Conference on the Theories and Practices of Securities and Financial Markets, the Seventh Biennial McGill Global Asset Management Conference in 2015, and the 2015 China International Conference in Finance for useful comments.

Endnotes

[1] The remaining reasons that trigger stock crashes include random shocks unexpected by the market. These reasons include, e.g., mergers and acquisitions, and options trading.

[2] An average investor cannot fully unravel the hidden information because of noisy inference. If the hidden information had been fully expected by investors, we would not have seen large share price reactions because share prices would have already priced the information anticipated by investors.

[3] Other factors impeding information disclosure include the uncertainty of investors about the endowment of the firms with the information (Dye 1985), the proprietary nature of the information (Darrough and Stoughton 1990), and so on. See Einhorn and Ziv (2012) for details.

[4] Specifically, the main competition variable, fluidity (which is constructed from financial statements), and all of the control variables obtained from Compustat are measured as of the fiscal year data. There are three control variables constructed using the stock market data, RET (returns), SIGMA (volatility), and DTURN (turnovers). Similar to the crash risk variables, we lag these three control variables by three months to align the accounting variable with the stock market variables. That is, these three stock-based control variables are measured over the 12-month period ending three months after a firm's fiscal year end.

[5] We also include industry-level returns (e.g., Fama–French 48 industry portfolio returns) besides overall market returns in Equation (1) and use the residuals to construct the crash risk measures. The resulting measure is thus more likely to be firm specific, with the industry-level factors removed. This better captures the firm-level competitive effect (where one firm in an industry faces more competitive threats than another firm in the same industry), rather than an industry-level effect (where there is more competition in one industry than another). We obtain robust results based on this measure and report them in Online Appendix Table A5.

[6] We also measure crash risk using 3.09 standard deviations (as in Hutton et al. 2009) rather than 3.2. 3.09 standard deviations will generate a crash frequency of 0.1%. We obtain similar results by using this alternative measure.

[7] The fluidity data are from 1997 to 2008 and are lagged in the regression analysis. As a result, our final sample period is 1998–2009.

[8] Our results hold if we control for industry-year paired fixed effects and for firm and year fixed effects. The results are reported in Online Appendix Table A6.

[9] Table 3 shows that the IV estimator is larger than the corresponding OLS estimator in magnitude. The gap between the two estimators may be caused by the downward bias in the OLS estimates attributable to measurement errors (Angrist and Krueger 2001). Our independent variable of interest, product market threats, is based on extracting words from financial statements and thus may be measured with errors. The second explanation is that OLS captures the overall average treatment effect (ATE), while the IV estimator captures the local average treatment effect (LATE). With heterogeneous treatment effects, LATE may differ from ATE (Card 2001, Angrist and Krueger 2001). For instance, firms with products more susceptible to competitions caused by tariff and exchange rate changes may have a larger effect on crash risk and thus IV estimates are larger. Angrist and Krueger (2001) emphasize that the local effects provide useful insights.

[10] We do not use industry deregulation as a shock to market competition because the literature finds that industry deregulation may be caused by industry performance and thus could be endogenous (Duso and Roller 2003).

[11] The mean import tariff rate is 1.1%, and the standard deviation is 1.9%.

[12] We choose the year 2009 for two reasons. First, 2009 is the year of the financial crisis and we can manually identify more stock crash cases. Second, 2009 is the last year of the sample because the competition fluidity data is only available up to that point.

[13] We find 43 stock crashes in the S&P 500 firms in 2009 but can only identify the trigger for 19 crashes through news searches.

[14] These random shocks relate to mergers and acquisitions, options trading, etc. For example, on November 1, 2009, one day after Denbury Resources released its acquisition of Encore Acquisition Co., Denbury's stock price dropped by 10%. In another case, Coventry Health experienced a significant price drop of 7% on September 15, 2009. The crash was due to the heavy selling pressure caused by the unusually high trading volume of the firm's put options.

[15] In Online Appendix Table A7, the subsample analysis conditional on institutional ownership shows that when a firm has high institutional ownership, the positive relation between competition and crash risk diminishes. The result confirms the role of monitoring and information asymmetry in affecting crash risk. Institutional investors can better monitor the managers and are more informed. Therefore, these investors can better anticipate the information hiding behavior (if any) and the adverse information is less likely to be "unexpected" for them. In addition, since firms with high institutional ownership are more transparent and have closer market monitoring, managers in such firms are less likely to conceal bad information and fewer crashes occur in these firms.

[16] Behavioral bias may contribute to this asymmetry in information gathering. For example, one manifestation of the overconfidence bias is that agents may think that good things are more likely, whereas bad events are less likely to happen to them (i.e., comparative-optimism effect; see Weinstein 1980). As a result, agents tend to collect good information rather than bad information.

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